

PENETRATION TESTING REPORT

Footprinting, Reconnaissance & Network Scanning Phases

W2-PM— CYBERSECURITY — NETWORKWALKS

Field	Detail
Pentester Name	Joel Biju
Program / Batch	B083-Networkwalks
Organization	Networkwalks
Week	Week 02
Date	13–19 September 2026
Modules Completed	W2-PM1 (Multiple Kali Tools), W2-PM2 (Google Dorking), W2-PM3 (Maltego), W2-PM4 (theHarvester OSINT), W2-PM5 (Zenmap Scanning)
Final Deliverable	W2-PM— Penetration Testing Report
Client / Target	1. Networkwalks (networkwalks.com) 2. Authorized local/lab network
Permission / Scope	Authorized educational cybersecurity training
Phases Covered	Phase 1: Reconnaissance & Footprinting Phase 2: OSINT Enumeration Phase 3: Scanning & Network Discovery

1 Liability Disclaimer

I performed these activities only within the authorized educational cybersecurity training scope and on systems/networks permitted for the practical exercises.

All materials in this report are provided for educational and research purposes only. The techniques demonstrated here should only be used on systems where appropriate authorization has been obtained or on systems owned by the tester.

Unauthorized access, scanning, enumeration, exploitation, or misuse of cybersecurity techniques may violate applicable laws and regulations.

No exploitation, credential attacks, persistence, privilege escalation, or destructive testing was performed as part of the activities documented in this report.

2 Introduction

This report documents the practical cybersecurity activities completed during **Week 2 of the Networkwalks Cybersecurity / Ethical Hacking Internship**.

The work covered five practical modules:

- **W2-PM1** — Footprinting & Reconnaissance using multiple Kali Linux tools
- **W2-PM2** — Google Dorking
- **W2-PM3** — Maltego
- **W2-PM4** — theHarvester-based OSINT enumeration
- **W2-PM5** — Zenmap / Nmap network scanning

The objective was to understand how a cybersecurity professional collects publicly observable information, identifies web technologies and DNS infrastructure, performs OSINT enumeration, discovers active hosts, identifies exposed services, and documents security-relevant observations.

The reconnaissance activities were performed using Kali Linux tools against the assigned training target, while Zenmap was used for an authorized local/lab network environment.

All findings in this report are treated as **observations rather than confirmed vulnerabilities** unless separately validated through authorized security testing.

3 Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux	Cybersecurity testing and reconnaissance environment
WHOIS	Find domain registration details, registrar, dates and name servers
WhatWeb	Fingerprint web technologies, CMS, plugins and server information
nslookup	Resolve the domain name to IP addresses using DNS
curl -i	Inspect HTTP response and security/application headers
Wafw00f	Identify the presence of a Web Application Firewall
DNSRecon	Enumerate DNS records including NS, MX, TXT, SOA and SRV
Exploit-db	Find GHDB dorks to footprint real targets through Google
Maltego	Gathering, connecting, and visualizing data to conduct cyber and open-source intelligence (OSINT)
theHarvester	Perform OSINT-based domain, host and infrastructure enumeration
Zenmap / Nmap	Discover live hosts and identify exposed TCP services
Local Network Tools	Support local network identification and scanning

4 Activities Performed

4.1 Footprinting & Reconnaissance

Reconnaissance was performed against the **networkwalks.com** domain using six Kali Linux tools: **WHOIS**, **WhatWeb**, **Nslookup**, **Curl**, **Wafw00f**, and **DNSRecon**. Each tool collected a specific category of information, including domain registration, web technologies, DNS details, HTTP metadata, WAF presence, and infrastructure records.

4.1.1 WHOIS

Purpose: Collect publicly available domain-registration information.

Command: `whois networkwalks.com`

Key Information Collected:

- **Domain:** NETWORKWALKS.COM
- **Registrar:** GoDaddy.com, LLC
- **Creation Date:** 2019-11-06 — **Registry Expiry:** 2027-11-06
- **Name Servers:** NS6135.HOSTGATOR.COM, NS6136.HOSTGATOR.COM
- **DNSSEC:** Unsigned
- **Domain Protection:** Delete, Renew, Transfer and Update prohibited

4.1.2 WhatWeb

Purpose: Fingerprint the target web application stack.

Command: `whatweb networkwalks.com`

Key Information Collected:

- **Web Server:** Apache — **CMS:** WordPress 7.0.4
- **Plugin:** WordPress Download Manager 3.3.58
- **JavaScript / Framework:** jQuery 3.7.1, Bootstrap
- **IP Address:** 192.232.216.135
- **Page Title:** Networkwalks Academy

4.1.3 Nslookup

Purpose: Resolve the target domain to its IP address via DNS.

Command: nslookup networkwalks.com

Key Information Collected:

- **IPv4 Address:** 192.232.216.135
- **IPv6 Address:** 64:ff9b::c0e8:d887
- **DNS Server:** 172.23.212.198

4.1.4 Curl

Purpose: Inspect HTTP response headers.

Command: curl -I https://networkwalks.com

Key Information Collected:

- **Status:** HTTP/2 200 — **Server:** Apache
- **REST API Endpoint:** /wp-json/
- **Cookie:** wpdm_client (HttpOnly)

4.1.5 Wafw00f

Purpose: Detect Web Application Firewall (WAF) presence.

Command: wafw00f networkwalks.com

Key Information Collected:

- **WAF Detected:** Yes
- **WAF Technology:** ModSecurity (SpiderLabs)

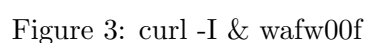
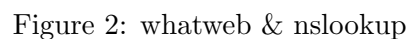
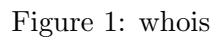
4.1.6 DNSRecon

Purpose: Enumerate DNS records.

Command: dnsrecon -d networkwalks.com

Key Information Collected:

- **SOA Server:** ns6135.hostgator.com
- **MX Record:** mail.networkwalks.com
- **SPF Record:** v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
- **SRV Record:** _autodiscover._tcp.networkwalks.com



Purpose: GHDB dorks to footprint real targets through Google

Task1 Find 10x live vulnerable security camera links that are exposed & accessible from Internet. List them in below table along with their relevant dork that you have used.

No.	Link	Relevant Dork	Username /Password (if any)
1	http://122.116.41.8:8080/	intitle:"webcamXP" inurl:8080	---
2	https://www.lmc.edu/webcam.htm	intitle:"Webcam" inurl:WebCam.htm	---
3	http://153.156.0.199/CAM/2023/03/	intitle:"Index of /cam/"	
4	https://ns.ph.liv.ac.uk/webcam/	intitle:"Index of /webcam/"	
5	https://meteo.physic.ut.ee/webcam/arhiiv/	intitle:"Index of /webcam/"	
6	https://www.skylinewebcams.com/en/webcam.html	inurl:webcam site:skylinewebcams.com inurl:roma	
7	http://109.233.191.130:8080/	inurl:webcam site:skylinewebcams.com inurl:roma	
8	http://www.insecam.org/en/bytype/webcamxp/#google_vignette	intitle:"webcamXP" inurl:8080	
9	https://klsimer.edu/img/infrastructure/lb/	intitle:"Index of" "DCIM/camera"	
10	http://208.107.246.94/control/camrainfo	inurl:control/camrainfo	

Figure 4: Live cams

Task2 Find 10x listings which contain downloadable mathematics ebooks in PDF format:

No.	Link	Relevant Dork	Username /Password (if any)
1	https://www.skylineuniversity.ac.ae/pdf/math/	intitle:index.of "parent directory" mathematics pdf	---
2	http://erewhon.superkuh.com/library/Math/	intitle:index.of "parent directory" mathematics pdf	
3	http://inis.jinr.ru/sl/vol2/Mathematics/Math.Encyclopedia/Pdf/	intitle:index.of "parent directory" mathematics pdf	
4	https://education.giakonda.org.uk/Maths/?SD	intitle:index.of "parent directory" mathematics pdf	
5	https://www.netlib.org/math/docpdf/	intitle:index.of "parent directory" mathematics pdf	
6	https://maths.nuigalway.ie/~rquinlan/linearalgebra/	intitle:index.of "parent directory" mathematics pdf	
7	https://ochicken.net/library/Mathematics/	intitle:index.of "parent directory" mathematics pdf	
8	https://www.maths.dur.ac.uk/papers/2025/	intitle:index.of "parent directory" mathematics pdf	
9	https://umeschandracollege.ac.in/pdf/study-material/mathematics-statistics/	intitle:index.of "parent directory" mathematics pdf	
10	https://www.nocities.org/amerommi@roners.com/cam/	intitle:index.of "parent"	

Figure 5: Mathematics pdfs

4.1.8 Maltego

Purpose: Conduct cyber and open-source intelligence (OSINT) investigations

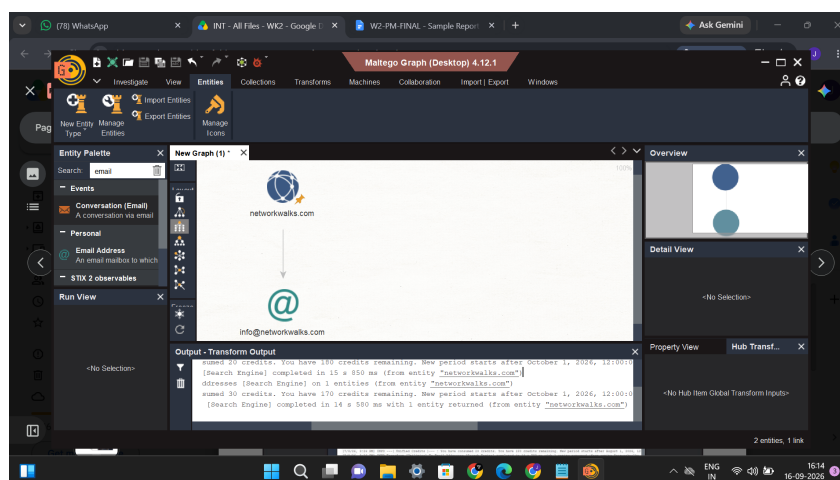


Figure 6: Maltego

4.2 OSINT Enumeration with theHarvester

Command: theHarvester -d networkwalks.com -l 10 -b all

Key Information Collected:

- **Target:** networkwalks.com
- **ASNs Found:** 3 (AS13335, AS31898, AS46606)
- **Interesting URLs:** 2 — **IP Addresses Found:** 4
- **Subdomains:** 3 (discovered via DNS fallback)

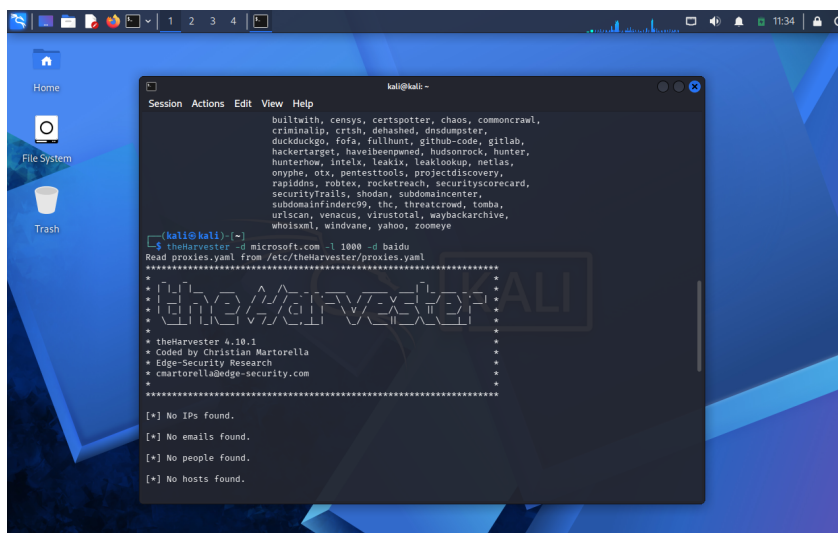


Figure 7: theHarvester

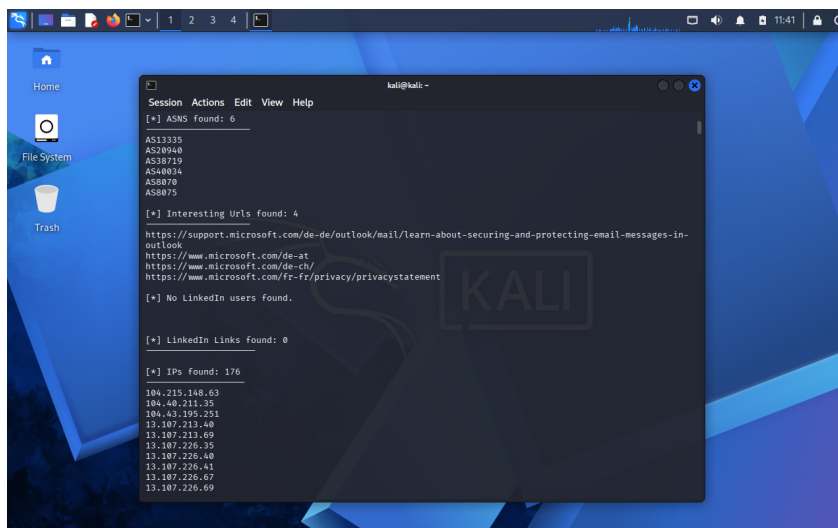


Figure 8: Findings

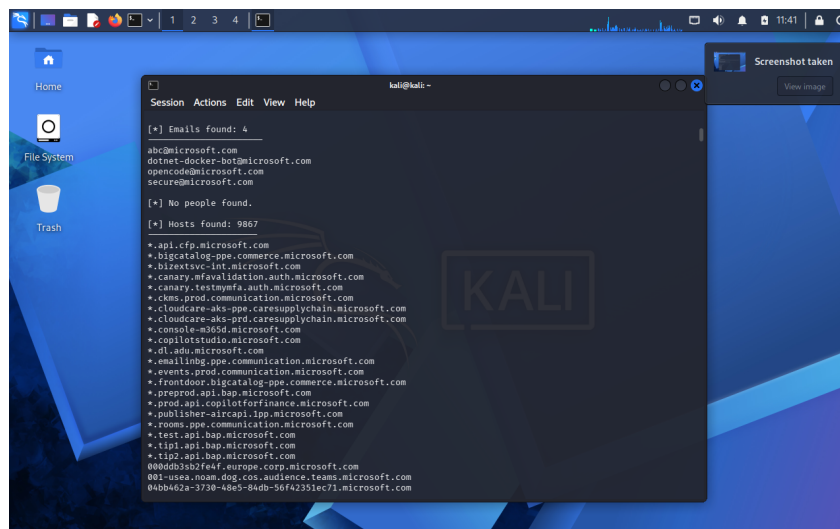


Figure 9: Findings

4.3 Network Scanning with Zenmap / Nmap

Target Range: 172.23.212.96/24 — **Command:** nmap -sn 10.0.0.2/24

The scan identified **2 active hosts**:

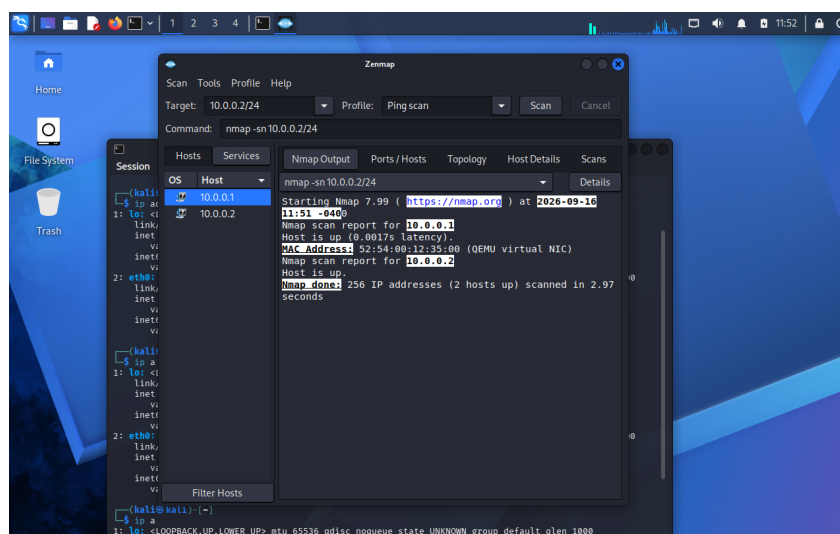


Figure 10: Zenmap

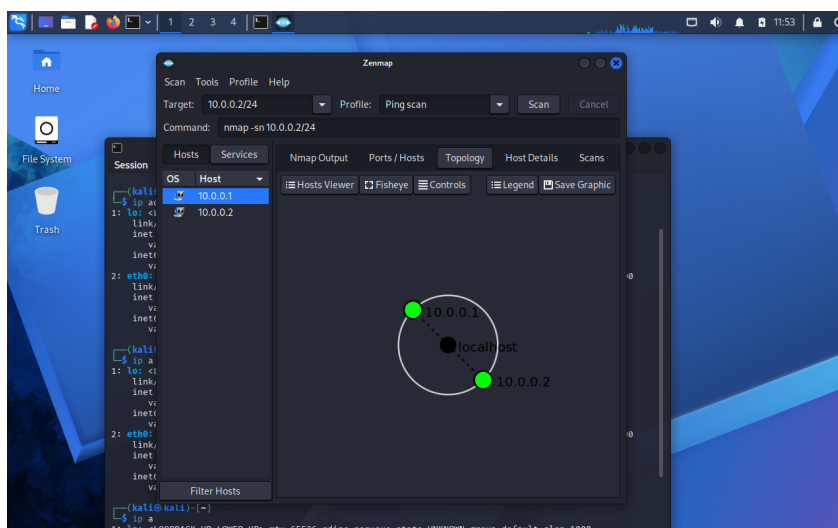


Figure 11: Live hosts

5 Risk Analysis / Impact

#	Risk / Finding	Potential Impact	Severity
1	Web technology info exposed	May assist technology fingerprinting by attackers	Medium
2	Public server IP identifiable	Reveals external network location	Low
3	HTTP headers exposed	Assists endpoint mapping (e.g. /wp-json/)	Low
4	WAF technology identified	Reveals security architecture details	Low
5	DNS records enumerated	Assists infrastructure profile building	Medium
6	OSINT info discoverable	Exposes ASNs, subdomains, and host IPs	Medium
7	Multiple active lab hosts	Increases target network attack surface	Medium
8	Exposed internal services	Services like SMB and PostgreSQL require strict access controls	Medium

6 Recommendations

1. **Minimize Information Disclosure:** Review and suppress unnecessary HTTP response headers and endpoint disclosures.
2. **Software Maintenance:** Keep WordPress core, plugins, and web servers regularly updated.
3. **DNS Security:** Periodically audit public DNS records and eliminate stale or unnecessary entries.
4. **WAF Configuration:** Ensure ModSecurity rule sets are updated and actively monitored.
5. **Internal Service Hardening:** Restrict internal services (e.g. SMB, PostgreSQL) using network firewalls and strong authentication.

7 Conclusion

During Week 2 of the internship, comprehensive reconnaissance, OSINT, and network scanning activities were carried out. Passive and active information gathering provided complete visibility into domain registrations, web application stacks, public DNS infrastructure, and internal network service exposures without executing destructive or intrusive testing.

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